

AROCKIA BILGATES

Full-Stack Software Engineer | MERN | Node.js | React | AI Applications

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Full-Stack Developer with 3+ years of experience building and deploying scalable MERN applications with AI integration. Specialized in embedding Gemini AI and real-time capabilities into production-grade web applications including AI coding platforms, AI-powered e-commerce systems, and live marketplace applications. Hands-on experience with REST APIs, JWT and OAuth authentication, Docker, CI/CD, and cloud deployment on AWS, Vercel, and Render. Seeking full-stack software engineering roles focused on scalable, AI-driven web products.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, Java, C++, SQL

Frontend: React.js, Next.js, Redux, React Hooks, HTML5, CSS3, Tailwind CSS, Bootstrap

Backend: Node.js, Express.js, Spring Boot, Django, Flask, REST API Development

Databases: MongoDB, MySQL, PostgreSQL, MongoDB Atlas, Firebase, Query Optimization

AI and ML: Gemini AI, OpenAI API, Judge0, scikit-learn, OpenCV, Machine Learning Fundamentals

Auth and Security: JWT, Google OAuth 2.0, bcrypt, RBAC, Role-Based Access Control

Real-Time: Socket.io, WebSockets, Server-Sent Events

DevOps and Tools: Git, GitHub, Docker, CI/CD, GitHub Actions, AWS (EC2, S3), Vercel, Render, Postman

Concepts: DSA, OOP, MVC, Microservices, System Design, Agile, SDLC

WORK EXPERIENCE

Freelance

Mar 2026 - Present

Full-Stack Developer | Remote / Coimbatore

- Architected and delivered production-grade MERN + AI client projects integrating Gemini AI APIs, Socket.io real-time features, and Razorpay payment gateways end-to-end.
- Built reusable REST API structures with JWT authentication, RBAC, and Docker-based deployment, reducing project setup time by 40% across client engagements.
- Designed and delivered cohort-based MERN and DSA programs, achieving 90%+ course completion rates.

Amypo Technologies Pvt. Ltd.

Jan 2025 - Feb 2026

Full-Stack Developer | Coimbatore, Tamil Nadu

- Developed and deployed production-grade MERN applications using React, Node.js, Express, and MongoDB with JWT authentication, RBAC security, and role-based dashboards.
- Architected scalable RESTful APIs following controller-service-repository layered patterns, improving maintainability and reducing average API latency by 30%.
- Implemented Docker-based CI/CD pipelines, reducing environment failures by 60% and enabling consistent production releases across 20+ engineering deployments.

Prag Robotics

Feb 2024 - Dec 2024

Python Developer | Thanjavur, Tamil Nadu

- Developed Python and OpenCV automation pipelines, improving processing speed by 30% and increasing ML-driven sensor accuracy from 85% to 94% via Flask REST services.
- Containerized microservices using Docker, enabling reproducible deployments and consistent real-time data exchange with robotic simulation environments.
- Refactored legacy automation scripts into modular service components, reducing code redundancy by 35% and improving maintainability.

CADD Centre Training Services

Feb 2023 - Dec 2023

Application Developer | Thanjavur, Tamil Nadu

- Built Java and Python backend systems with modular architecture, REST API integrations, and database-driven business logic modules.
- Applied OOP principles and Git version control best practices across full SDLC including requirement analysis, development, testing, and deployment.

Freelance

May 2021 - Jan 2023

Full-Stack Web Developer | Remote

- Delivered full-stack web applications for clients using React, Node.js, Express, and MongoDB with end-to-end ownership from requirements to production deployment.
- Developed REST APIs with JWT authentication, integrated third-party services, and built responsive mobile-first React frontends independently.

PROJECTS

PrepGenius - AI Coding Interview Preparation Platform

Tech: React, Node.js, Express, MongoDB, Gemini AI, Judge0, Monaco Editor, Tailwind CSS

GitHub: github.com/Bilgates567/prepgenius

- Built a MERN coding platform with Monaco Editor and Judge0 automated judging for 20 DSA problems, enabling browser-based code execution with instant verdict feedback.
- Integrated Gemini AI as a post-submission mentor streaming time/space complexity analysis and personalized interview coaching via Server-Sent Events.
- Implemented a 3-level progressive hint system, real-time AI chat assistant, and personalized weekly skill roadmap simulating a complete interview coaching environment.

AroCart AI - AI-Powered E-Commerce Platform

Tech: React, Node.js, Express, MongoDB, Gemini AI, Razorpay, Google OAuth, Tailwind CSS

GitHub: github.com/Bilgates567/arocart-ai

- Built Gemini-powered smart search with intent detection, AI shopping assistant, and AI product comparison engine enabling context-aware product discovery.
- Integrated Razorpay (UPI/Cards/COD) with HMAC SHA256 payment verification, Google OAuth, and 6-stage live order tracking across 64 products in 6 categories.
- Designed scalable REST APIs with input validation, structured error handling, and optimized MongoDB indexing for fast product retrieval under load.

QuickHelp - Real-Time Micro-Task Marketplace

Tech: React, Node.js, Express, MongoDB, Socket.io, Google OAuth, Tailwind CSS

GitHub: github.com/Bilgates567/quickhelp

- Built a two-sided MERN marketplace with Socket.io real-time bid notifications, live status updates, and helper online/offline tracking across a full task lifecycle.
- Implemented RBAC-secured role-based dashboards for users, helpers, and admin moderation with dispute resolution and full audit trails on task events.

EDUCATION

Bachelor of Engineering in Computer Science Engineering

Jun 2017 - Apr 2021

St. Joseph's College of Engineering and Technology | CGPA: 7.21

KEY ACHIEVEMENTS

- Built AI-integrated MERN applications leveraging Gemini AI, Socket.io, Judge0, and role-based backend architectures.
- Developed scalable REST APIs with JWT authentication, RBAC, Docker deployment, and cloud-native workflows.
- Engineered real-time features including live notifications, AI chat systems, and collaborative coding experiences.
- Delivered production-grade full-stack solutions across e-commerce, interview preparation, and marketplace platforms.